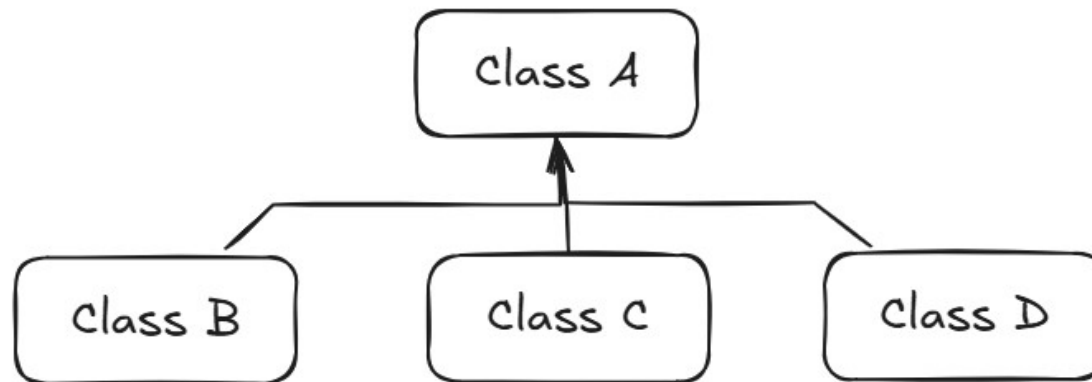


✓ Hierarchical Inheritance:

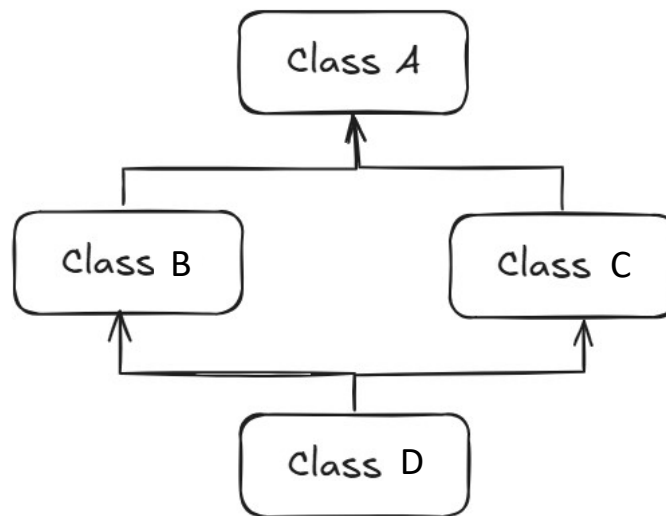
- Hierarchical inheritance is when multiple child classes inherit from a single parent class.
- This means **one parent, many children** — each child gets attributes and methods of the same parent but may also have its own attributes and methods.



✓ Hybrid Inheritance:

Hybrid inheritance is a **combination of two or more types of inheritance** (single, multilevel, multiple, hierarchical) in the same program.

It often creates a **complex inheritance structure** where one class is derived in more than one way.



3. Polymorphism: Poly means “many forms”

Polymorphism is a way in which different classes can share the same method name, but those methods can act differently based on which objects call them.

In Polymorphism, the methods of classes i.e. Parent/s or child/s are same, but the underlying execution differs depending on the object's class.

4. Abstraction:

Abstraction means hiding implementation details and showing only the essential features to the users. In Python, Abstraction is implemented using abstract classes and abstract methods.

The concept of hiding the complex reality while exposing only the essential functionality or interfaces to the users. It helps reduce programming complexity and efforts.

We cannot create the object of abstract class directly. It's purpose is to be directly inherited by other classes.

Abstract classes defines a set of methods that subclasses are required to implement (Blueprint for classes)